GlucoTwin — system architecture

A patient-specific type 2 diabetes digital twin. Four input streams are harmonised onto one causal timeline, fused by the model layer, calibrated into prediction intervals and adverse-event probabilities, and served to two clinician interfaces. Raw patient records never leave the clinic.

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1 · SOURCES

Inside the clinic — raw records never leave

STATIC EHR

Health record

Age, sex, BMI, HbA1c, eGFR, diagnoses, medications, genetic markers. Synthea-ready.

REAL · OPEN

CGM · 145 patients

CGMacros (45 adults) and Shanghai T2DM (100). Removing it hurts the forecast most.

REAL + SIMULATED

Wearable

Fitbit HR and activity; sleep, HRV, steps simulated. At +60: **+0.04 mg/dL, CI spans zero**.

REAL + PRESETS

Meals

Logged macronutrients, plus 16 Indian presets from IFCT-2017. Meals are the **largest single gain**.

2 · HARMONISE & FUSE

One causal timeline

Common grid. Every cohort resampled to a single 5-minute schema with a shared column set; loader checksums verified.

Causal features only. No feature may use information from after the prediction instant — enforced in code and covered by tests.

Patient-grouped splits. Training, inner-validation, calibration and holdout patients never overlap, so no patient appears on both sides of an evaluation.

Gap and quality handling. Sensor gaps flagged rather than silently interpolated; a per-window quality score travels with the forecast.

Fused features. CGM history and slope · time of day · meals · heart rate and activity · static profile.

`src/data · src/features`

3 · MODEL LAYER

Fusion forecaster

Tuned LightGBM on the *change* from the current value, at three horizons. The served model.

Mechanistic twin

A per-patient glucose–insulin ODE: RK4 integration, extended Kalman filter state.

Hybrid physics + ML

Twin features fed to the forecaster; a GRU trained with an ODE-consistency loss.

Personalisation

Fine-tuning or recursive least squares: ≈1 mg/dL better after 3 to 5 of the patient's own days.

Federated training

FedAvg and FedProx across simulated clinics; five-clinic and two-site layouts.

4 · CALIBRATE

80% intervals

Split-conformal and CV+. Coverage 80.8% (CGMacros) and 79.5% (Shanghai) at +60 min, per subgroup as well as pooled.

Event probabilities

Platt and isotonic calibration for crossing 180, 140 and 70 mg/dL. Calibrated means 40% happens about 40% of the time.

Robustness & quality

Faults injected, features recomputed, scored on the same anchors. Quality: completeness, recency, plausibility.

Alerts & explainability

Alert-burden curves trade sensitivity against false alarms explicitly. SHAP explains each forecast.

5 · CLINICAL OUTPUTS

Forecast + band

Trajectory to +120 min with its 80% interval and one plain sentence. The band widens with the horizon — that widening is the content.

Adverse-event risk

Calibrated chance of hyper- or hypoglycaemia in the window, at the clinic's chosen threshold (2019 consensus).

What-if answers

Counterfactual meal size, carbohydrate, timing, an Indian preset, and a post-meal walk.

Trust surface

Sensor data quality, personalisation depth, input tier available, feature attribution, and the limits printed beside each answer.

6 · INTERFACES

Web console `web/`

A single-page clinician console. Patient list with triage ordering, then per patient: forecast, what-if lab, glucose report (AGP, time in range, GMI per the 2019 consensus), health record, sleep and activity context, visit summary, and an evidence section carrying every cohort-level result.

Role-gated access — clinician and care coordinator see different fields — with a session access log. Dark and light themes, mg/dL ⇄ mmol/L, readable at tablet width.

Streamlit application `make app`

Ten pages: Overview, Live twin, Glucose report, What-if lab, Twin state, Personalization, Input tiers, Data quality, Evidence, and a PDF export for the patient file.

Both interfaces read the same bundle, so the two never disagree on a number. Every page carries the decision-support disclaimer: the clinician remains the decision maker and the system never recommends a dose or a drug.

METHOD & EVALUATION

- **Pre-registered.** Protocol, candidates and decision rule written to `docs/DECISIONS.md` before the run. 70+ entries.
- **Patient-grouped CV** with repeats, an inner split for every choice, and a holdout the runner refuses to reuse.
- **Baselines it must beat:** persistence, per-patient autoregression, ridge, a GRU.
- **Paired patient bootstrap** — intervals over patients, never over time windows.
- **Also scored:** coverage, event AUROC, alert burden, plausibility, 87 matched breakfasts, input-tier ablations; stressed by transfer, faults and a synthetic Indian cohort.
- **Powered simulation (sleep).** 777,000 scored windows: at literature effect size the sleep channel is worth ≈0.06 mg/dL — about **9,000 patient-days before it is visible at all**. Nulls published at full size.

AI / ML MODELS AND TRAINING

- **Inputs → target.** Causal features per anchor: CGM lags, rolling statistics, slope and acceleration, time-of-day sin/cos, meal macros, heart rate and activity, static profile. Target is the *change* from the current value.
- **Served forecaster.** LightGBM, one head per horizon, trained with an absolute-error loss to match the metric it is scored on. 16 candidates searched, best three averaged, on inner-validation patients only.
- **Sequence model.** PyTorch GRU over the causal window with FiLM static conditioning; also the backbone of the physics-informed hybrid.
- **Constrained what-if.** A second LightGBM with monotonicity constraints: less carbohydrate can never predict a higher peak, by construction.
- **Uncertainty.** Split-conformal and CV+ intervals; Platt and isotonic probability calibration.

MEASURED RESULTS

METRIC	VALUE
MAE at +30 min	11.6
MAE at +60 min	16.8
MAE at +120 min	21.2
AUROC, above 180	0.82
AUROC, below 70	0.81
Interval coverage	80.8%
Transfer gap	3.4

PRIVACY BY DESIGN

- Anonymised open or synthetic data only; no real patient record is in the repo.
- **Federated:** FedAvg matches pooled training (0.09–0.12 mg/dL, intervals include zero) and beats a lone small clinic by ≈2 mg/dL.
- Data minimisation by design; no formal differential-privacy guarantee is claimed at this clinic count.
- Role-gated access with a session log; mapped to India's **DPDP Act** in `docs/limitations.md`.

ENGINEERING & SCOPE

- Python 3.11 · LightGBM · PyTorch · SHAP · Streamlit. Laptop CPU; no GPU.
- One `make` target per stage; every figure and table regenerates from source. CI runs ruff, black, pytest and a smoke run. Docker image, datasheet and model card included.
- One condition: **type 2 diabetes**. Decision support only — not a medical device.
- India, sleep and federated results are **simulations**: they test design, not clinical performance. Not prospectively validated.

Code MIT. Datasets keep their own licences. Simulated streams are labelled SIMULATED throughout.